

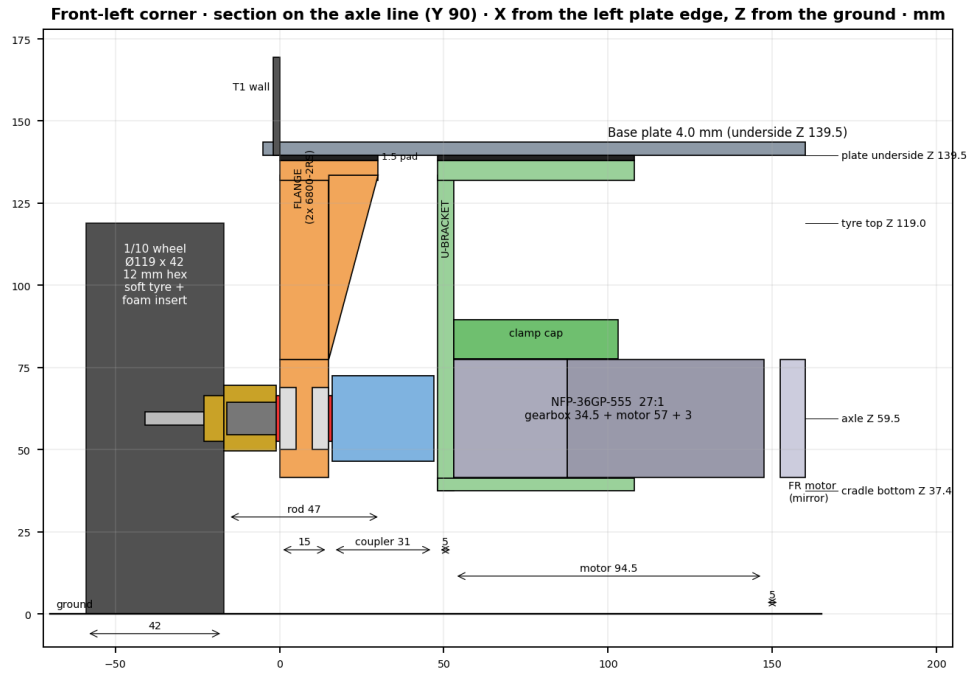
Mine Rescue Rover · 4WD Drivetrain (under the base plate)

Four identical corner units (mirrored left/right): wheel -> machined hex drive adaptor -> 10 mm steel stub axle -> 3D-printed bearing flange (2 bearings) -> 3D-printed coupler -> 36 mm planetary gear motor held in a 3D-printed U-bracket. No encoders (the LIDAR 2D SLAM gives odometry). Axles run parallel to the 30 cm side on Y = 90 (front) and Y = 310 (rear). Every dimension below comes from one model that is checked by script (results on the last page: **0 failures**).

Design decisions and fixes

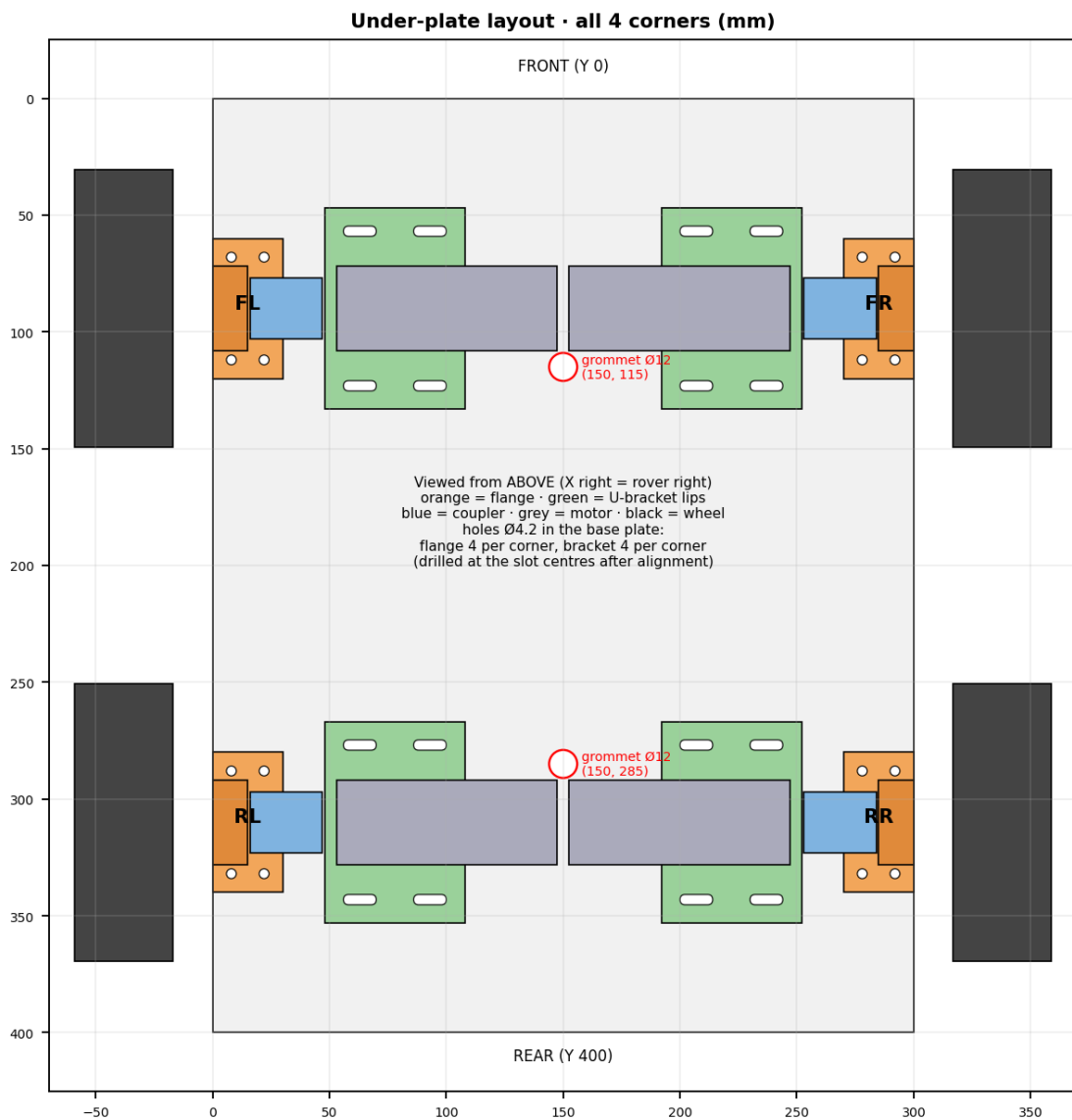
#	Old spec	This design	Reason
1	Motor + encoder 90 mm; L-bracket at X 30; void 6 cm	No encoder. NFP-36GP-555, 27:1 (gearbox 34.5 mm); motor face at X 53, rear at X 147.5; 5 mm between opposite motors	A 555 planetary motor is ~95 mm long. Two face-to-face on a 300 mm plate only fit with the short 27:1 gearbox (51-139:1 gearboxes are 42.5 mm and collide)
2	Rod 50 = 15 / 15 / 20	Rod 47 = 15 adaptor / 1 shim / 15 bearings / 1 shim / 15 coupler	Coupler engagement cut 20 -> 15 to make room for the motors; shims keep spinning parts off the printed flange
3	Standoff block + zip tie through the plate	U-bracket cradle + clamp cap hold the motor along 60 mm	Removes the cantilever without holes under the batteries
4	Wheel Ø120	1/10 RC wheel Ø119 x 42, 12 mm hex (Stampede-type set)	Real, stocked part with the right hex drive; axle Z 59.5, drop still exactly 80.0
5	Motor leads through a 6 cm central void	2 grommets Ø12 at (150, 115) and (150, 285) between the motors	Clear of the LiPo, 12 V bus, 5 V bus and Battery B above
7	Rigid wheel, no damping	Soft-compound tyre with open-cell foam insert, Ø119 x 42, pre-glued to a 12 mm hex rim	The tyre squashes ~3 mm under load and absorbs shock before it reaches the gearbox and printed parts
8	Flange bolted straight to the plate	1.5 mm neoprene pad (60 Shore A) under the flange AND under the bracket lips; both printed 1.5 mm shorter	Damps vibration. Both get a pad so the axle and motor stay exactly coaxial (the coupler is rigid)
9	Bolt heads on the base plate top	M4x20 countersunk (DIN 7991), heads flush in 90° countersinks	Nothing pokes up under Battery C, the regulator or the Tier 1 mat
6	Motor L-bracket at X 30 (fixed)	Bracket lips have ±5 mm slots; plate holes drilled at the slot centres after alignment	Makes the build independent of small motor tolerances (match-drill protocol)

Sheet 1 · Corner geometry



Part	X from - to (left corner; right = 300 - X)	Z / section
Wheel (tyre)	-59.0 to -17.0	Ø119, centre Z 59.5, top Z 119.0 (20.5 below the plate)
Hex adaptor	stud -41 to -23 · hex -23 to -17 · collar -17 to -1	collar Ø20, hex 12 A/F
Shim (outer)	-1 to 0	10 x 14 x 1 steel shim, rides on the bearing inner race
Flange boss	0 to 15 (outer face flush with the plate edge)	Ø36, 2 x 6800-2RS
Flange top plate	0 to 30, Y ±30 about the axle	6 thick, Z 132.0-138.0, pad Z 138.0-139.5
Shim (inner)	15 to 16	
Stub axle	-16 to 31	Ø10 x 47
Coupler	16 to 47	Ø26
U-bracket front wall	48 to 53	shaft hole Ø14
U-bracket cradle / lips	48 to 108	lips Y ±23.1 to ±43.1 about the axle, Z 132.0-138.0, pad above
Gearbox / motor	53 to 87.5 / 87.5 to 147.5	Ø36, Z 41.5-77.5
Clamp cap	53 to 103	Z 77.6-89.6
Gap to the opposite motor	147.5 to 152.5	5 mm

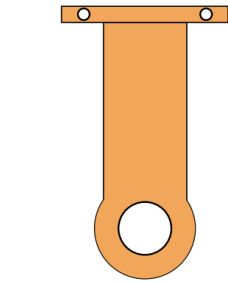
Sheet 2 · Under-plate layout



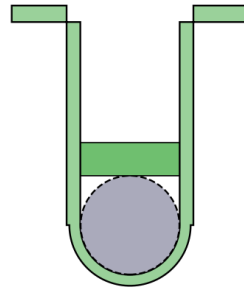
Base plate holes per corner (left front shown, mirror in X for right, Y -> 310 for rear): flange 4 x Ø4.2 with 90° countersink to Ø8.5 on the TOP face, at X 8 and 22, Y 68 and 112. Bracket 4 x Ø4.2 (same countersink) at the slot centres, nominal X 63 and 93, Y 56.9 and 123.1, drilled after the alignment step. Right-side X = 300 - X.

Sheet 3 · Parts

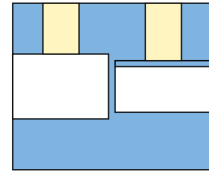
Printed and machined parts (mm)



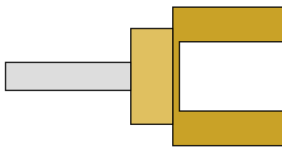
FLANGE - view along the axle
boss Ø36 x 15 - pockets Ø19.0 x 5 both faces
centre bore Ø11 - web 30 wide x 15 thick
top plate 60 x 30 x 6, 4 holes Ø4.4 at Y ±22, X 8/22
2 gussets 15 x 15 on the inboard face



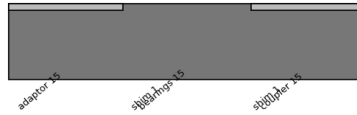
U-BRACKET - end view (from the wheel)
cradle R18.1 inside, 4 mm under the motor
arms 5 mm, top of arms flush to plate
lips 20 x 6 each side, 60 long, 2 slots each
front wall 5 mm: shaft hole Ø14 + motor M3 holes



COUPLER - section
Ø26 x 31, 100 % infill
rod bore Ø10.1 x 15 - gap 1
shaft bore Ø8.1 D-flat (7.1) x 15
2 x M4 inserts per side at 90°, M4x6 grubs



HEX DRIVE ADAPTOR - turned 6061-T6 (or EN8)
collar Ø20 x 16, bore Ø10 H7 x 15 deep
2 x M5 grub at 90° onto rod flats
hex 12 A/F x 6 (seats in the wheel hex)
stud M4 x 18 + M4 flanged nyloc (wheel nut)



STUB AXLE - EN8 bright bar Ø10 h9 x 47
0.5 mm chamfers; two flats 0.8 deep x 13 long at 90°
on each end (grub-screw seats)

CLAMP CAP
60 long (X) x 36.2 wide x 12 tall
underside R18.1 saddle + 1 mm rubber strip
4 x M4 heat-set inserts in its side faces
(2 per side), M4x20 through the arms

Replaces the wooden standoff block and
zip tie: the cradle carries the motor
from below, the cap clamps it from above.

3D-printing settings

Part	Qty	Material	Orientation on the bed	Settings
Flange	4	PETG (upgrade: PA6-CF or PETG-CF, hardened nozzle)	Outer boss face (X 0) down: the web and boss print as in-plane layers, bearing pockets print as true circles	0.4 nozzle, 0.2 layer, 6 walls, 6 top / 6 bottom, 60 % gyroid, 240 °C / bed 80 °C, supports only under the top-plate bolt holes
U-bracket	4	same	Front wall down: every layer is the full U profile, so arm bending stays in-plane	same, 60 % gyroid, slots printed
Clamp cap	4	same	Flat top down	5 walls, 40 % gyroid; press 4 M4 heat-set inserts after printing
Coupler	4	PETG or PA6-CF	Axis vertical, rod bore up	100 % infill (rectilinear), 0.15 layer; 4 M4 heat-set inserts
Test ring	1	same	flat	Print a Ø19.0 pocket ring first: the bearing must press in by hand-vise. If loose, retain with Loctite 641 (or thin CA)

Printed holes are drawn at 4.4 mm (not 4.2) because printed holes shrink about 0.2 mm; drill to 4.2-4.4 if tight. Metal plate holes stay 4.2 mm.

Sheet 4 · Procurement (for 4 corners)

Item	Exact part	Qty	Note
Motor	NFP-36GP-555 12 V, 27:1 (gearbox 34.5 mm), Ø8 D-shaft	4	Before buying, confirm the listing's shaft extension (boss + shaft). Cut it to 21 mm if longer; 17 mm minimum. Match-drill the M3 face holes from the gearbox
Bearings	6800-2RS, 10 x 19 x 5	8	2 per flange
Stub axles	EN8 (or C45) bright bar Ø10 h9, cut to 47	4	Mill or file the grub flats (sheet 3)
Hex adaptors	Turned to the sheet 3 drawing, 6061-T6 or EN8	4	Any local lathe shop; bore Ø10 H7
Wheels	1/10 monster-truck wheel + tyre, 12 mm hex, Ø119 x 42, soft compound, open-cell foam insert, pre-glued	1 set of 4	Choose "soft" compound. The foam insert gives the cushioning; do not fit closed-cell or dual-stage hard inserts
Pads	Neoprene (CR) sheet 1.5 mm, 60 Shore A	8 pads	Cut 60 x 30 (flange) and 60 x 20 x 2 (lips), holes Ø4.5
Shims	10 x 14 x 1 steel shim washers	8	
Bolts	M4 x 20 countersunk DIN 7991 + 1 mm rubber washers + M4 nyloc	32 sets	16 flange + 16 bracket lip, heads flush with the plate top
Cap bolts	M4 x 20 socket head	16	Into the cap inserts, no nut
Heat-set inserts	M4, 5.6-5.7 mm OD, 8 mm long (cap) / 6.4 mm (coupler)	16 + 16	
Grub screws	M4 x 6 (coupler) · M5 x 6 (adaptor), cup point	16 · 8	Loctite 243
Wheel nuts	M4 flanged nyloc	4	
Motor screws	M3, length = bracket wall 5 + thread depth in gearbox (usually 3-4) -> M3 x 8	16-24	
Grommets	rubber, 12 mm hole	2	
Filament	PETG (or PA6-CF / PETG-CF)	~1 kg	

Sheet 5 · Assembly (base plate upside down on the bench)

1 · Prepare

Print all parts and the test ring. Press the heat-set inserts. Cut and flat the axles. Deburr. Drill the flange holes and grommets in the base plate (sheet 2); leave the bracket holes for step 6.

2 · Flanges

Press two 6800-2RS into each flange (one per face, flush). Lay the 1.5 mm neoprene pad on the flange top, then bolt it to the base plate: countersunk head flush in the plate top, then plate, pad, flange, rubber washer, nyloc underneath. Tighten until the pad just starts to bulge (do not crush it). The boss outer face must be flush with the plate edge.

3 · Axles

Slide the outer shim, then the axle through the bearings from outside, so 15 mm protrudes outboard beyond the shim and 16 mm inboard. Fit the inner shim.

4 · Couplers

Push the coupler rod bore fully onto the inner 15 mm of the axle, against the inner shim (the shims should turn freely, with no end play). Tighten both grubs onto the flats (Loctite 243).

5 · Motors

Screw each motor face to the U-bracket front wall (M3). Lay it in the cradle with the rubber strip, fit the clamp cap and tighten the 4 cap bolts.

6 · Align and drill (match-drill)

Slide the motor shaft into the coupler until the bracket lips sit flat on the plate and the wall is 1 mm from the coupler end. Mark the plate through the 4 slot centres, remove the motor, drill Ø4.2 and countersink the top, then refit with the lip pads and bolt (same stack as the flange, same pad squeeze). Tighten the coupler shaft grubs onto the D-flat. Turn the wheel end by hand; it must turn with no binding.

7 · Wheels

Slide the hex adaptor onto the outer 15 mm (against the outer shim), tighten both M5 grubs onto the flats (Loctite 243), fit the wheel onto the hex and tighten the M4 flanged nyloc.

8 · Wiring

Motor leads (red/black) go up through the grommets: FL + RL to Driver 1 M+ / M- and FR + RR to Driver 2 (Part 1 wiring). If one side drives backwards, swap M+ and M- at that driver.

9 · Check

Put the rover on the floor: tyres carry the load and squash about 3 mm, the lowest point (cradle) is about 34.5 mm up, and the gap between opposite motors is 5 mm. Run each side at low PWM and listen for rubbing.

Sheet 6 · Verification

Check (script)
OK Flange/bracket drop plate-underside to axle = 80.0 mm (spec 80.0)
OK Axle height = tyre radius = 59.5 mm
OK Tyre top 119.0, plate underside 139.5 -> clearance 20.5 mm
OK Rod 47 mm = adaptor bore 15 + flange bearings 15 + coupler 15 + 2 shims
OK Coupler end 47.0, bracket wall starts 48.0 (gap 1.0)
OK Motor shaft engagement in coupler 15.0 mm (min 12)
OK Coupler clears flange boss (shim 1.0 mm between)
OK Left motor rear 147.5, right motor rear 152.5 -> gap between motors 5.0 mm
OK Flange top plate ends X 30, bracket lips start X 48
OK With 3 mm soft-tyre squash under load: lowest point 34.5 mm
OK Lowest point (bracket cradle) 37.5 mm above ground; motor 41.5; flange boss 41.5
OK Tyre inner face at X -17.0 -> 17.0 mm outside the plate edge / flange
OK M4x20 countersunk flange (CSK flush) + 1.5 pad: plate 4 + pad 1.5 + part 6 + rubber washer 1 + nyloc 5 -> 2.5 mm past nut, head flush with the plate top
OK M4x20 countersunk bracket lip (CSK flush) + 1.5 pad: plate 4 + pad 1.5 + part 6 + rubber washer 1 + nyloc 5 -> 2.5 mm past nut, head flush with the plate top

Check (script)
OK M4x20 lower T1 bracket (chassis doc): 7.2 mm past nut
OK Pads: flange and bracket lips both on 1.5 mm pads and both printed 1.5 mm shorter -> axle and motor stay coaxial at 80.0 below the plate
OK M4x20 arm into cap insert: 14.2 mm of thread in the cap (insert 8.1 long, hole 16 deep)
OK Under-plate footprints: flange top, bracket lips, chassis bolt heads, grommets - no overlaps
TOTAL FAILURES: 0

Confirm on the real parts

#	Check
1	Motor shaft extension (face to tip) is 17-21 mm; cut to 21 if longer. With 21 the coupler takes 15 mm of shaft.
2	Motor body length (gearbox + can + rear boss) is 94.5 mm or less, otherwise the opposite motors touch. Measure before drilling.
3	Wheel: the hex seat is 12 mm and the rim inner face sits on it (the tyre inner face ends up 17 mm outside the plate edge).
4	Bearing pocket press fit on the test ring before printing all four flanges.
5	After fitting the pads, check the axle and motor turn freely by hand; if the coupler binds, the two pads are squeezed unequally.

Sheet 7 · Insulation mat (chassis, all decks)

A mat on every enclosed deck floor keeps loose wires, solder joints and bolt tips off the aluminium. It goes **inside each deck only**, not on the exposed ledges of the lower roofs and not on the top roof.

Item	Specification
Material	Silicone rubber sheet, 1.5 mm, 50 Shore A, flame-retardant grade (UL94 V-0), rated to 200 °C. Silicone survives the heat near the BTS7960 heatsinks and does not melt onto the plate
Tier 1	30.0 x 40.0 cm on the base plate
Tier 2	22.0 x 35.0 cm on the mid plate, inside the Tier 2 box
Tier 3	18.1 x 31.1 cm on the Tier 2 roof, inside the Tier 3 box
Cut-outs (+1 mm all round)	Pillars, every lip / lower bracket, latch posts and the wire pass-through holes: brackets must clamp metal to metal, never on the mat
Cut-outs under breadboards	All 400-pt and 170-pt breadboards stick straight onto the aluminium (their adhesive will not hold on silicone). This keeps the MQ and MLX90640 heights, so the gas ports (4.40 cm) and thermal arches (3.5 cm) are unchanged
Holes	Punch Ø5 for every bolt and standoff that passes through the floor (drivers, bus bars, Pi 5, hub, battery bays)
Fixing	Held down by the parts bolted through it; add a few dots of neutral-cure RTV silicone at the corners
Height effect	Parts sitting on the mat rise 1.5 mm. The tallest (Tier 1 drivers, 4.5 cm) become 4.65 cm, still 2.8 cm below the nearest bolt tip hanging from the mid plate (7.42 cm). All drivetrain bolt heads are countersunk flush, so nothing pushes up under the mat

Quantity: one 50 x 50 cm sheet covers Tier 1; one more sheet covers Tier 2 and Tier 3.